

HAT-P-32b

R-0230

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_251207 · created 2026-07-17T03:30:12+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461016.6712 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1526 +/- 0.0083
Transit depth (Rp/Rs)^2	2.33 +/- 0.25 [%]
Orbital Inclination (inc)	89.99 +/- 1.33
Airmass coefficient 1 (a1)	1.0148 +/- 0.0063
Airmass coefficient 2 (a2)	-0.014 +/- 0.048
Scatter in the residuals of the lightcurve fit is	0.62 %
Best Comparison Star	None
Optimal Aperture	3.8
Optimal Annulus	7.87
Transit Duration (day)	0.1099 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1526 ± 0.0083 vs 0.1489 (deviation 0.41σ)
Duration fit vs published	0.1099 d vs 0.1304 d (deviation 7.46σ)
Mid-time O–C	-1.1 min
Depth SNR	16.7
Residual scatter	0.62 %

Light curve

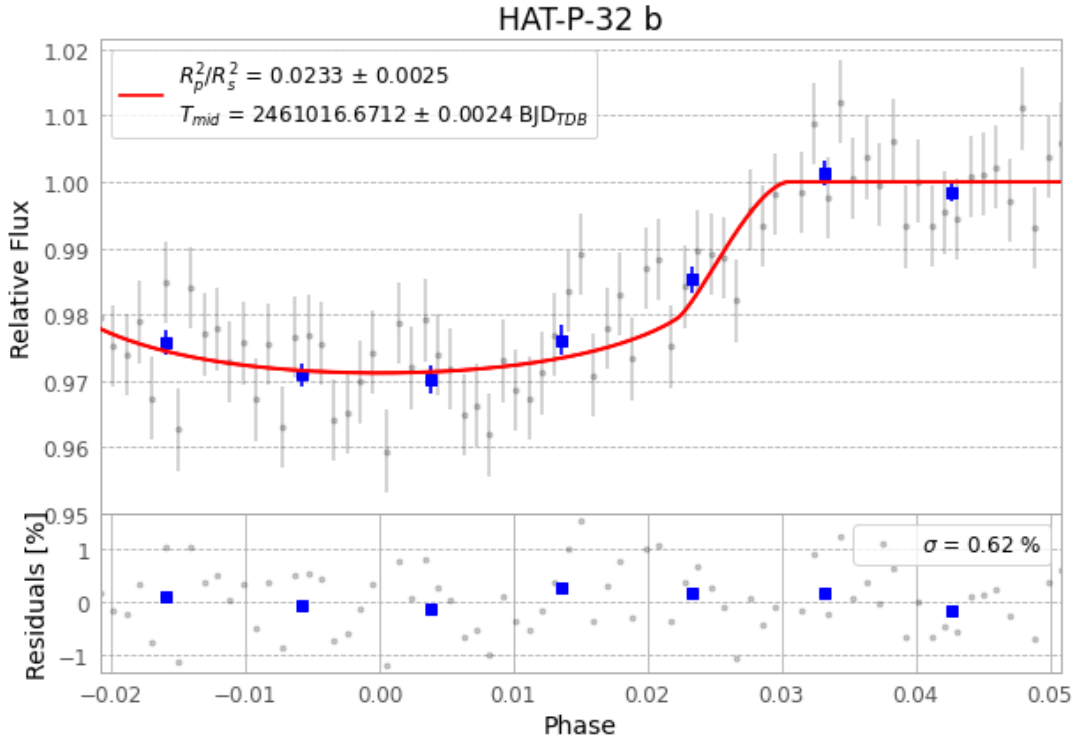


Figure 1 — FinalLightCurve_HAT-P-32 b_2025-12-06.png (SHA-256 4cd1893dbe98a60b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 316], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17360ebcacf4d29d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

75 FITS frames (49 MB), HATP-32251207025416.FITS → HATP-32251207063615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-251207.log (f3d9eaf91f4b...), FinalLightCurve_HAT-P-32 b_2025-12-06.png (4cd1893dbe98...), FinalLightCurve_HAT-P-32 b_2025-12-06.csv (8f4539977131...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 03:30Z.